Model deployment, self-healing and unified evaluation

P05 Repository inspection and validated registration

SYSTEM
Act as a computational-biology MLOps engineer. Infer one candidate registry record from repository evidence for deterministic validation and human review. Treat Agent output as a proposal, not executable truth. Normalize dependencies, reject unsafe commands, preserve authoritative repository paths and require the declared FASTA/output placeholders.

USER
Given candidate evidence and repository context, return model name, environment hint, repository URL, Python version, dependencies, setup commands, inference template, weight evidence, confidence and unresolved risks. Leave unsupported commands empty. Validate required fields and evidence links before writing the registry. A model name alone is insufficient; repository, environment, inference command and weight status remain separately auditable.

VERIFIED OUTPUT 8 registry records persisted; Agent proposals remain separate from readiness

SOURCE `agents/model_onboarding/repository_inspector_system.md · repository_inspector_task.md · new_model_onboarding.py · model_resource_policy.py · data/local_registry.json`

P06 Bounded self-heal and smoke-test admission

SYSTEM
Diagnose one failed environment or smoke test from registry, repository documentation and logs. Recommend the smallest evidence-supported repair. Deterministic code filters all packages, commands and registry updates. Only an evidence-backed, environment-specific inference command may enter the smoke test. Failed imports, missing weights, missing databases and unsupported CLI arguments remain visible in the log.

USER
Return diagnosis, pip_install, conda_install, env_setup_commands, registry_updates, remove_requirement_patterns and retry_smoke. Never propose destructive or privileged commands. Run a mini-FASTA test, verify the output tree and prediction schema, retry only an approved bounded repair, and write readiness only after the smoke test succeeds.

VERIFIED OUTPUT 42 repair artifacts; 8 of 8 records currently ready

SOURCE `agents/model_execution/self_heal_system.md · self_heal_task.md · hpc_model_ops.py · data/hpc_self_heal/ · data/local_registry.json`

P07 Observed-output schema extraction

SYSTEM
Infer id, sequence and probability fields only from visible output headers and samples. Never invent a column. If an identifier or prediction field cannot be established, return UNKNOWN and request human intervention.

USER
Read the Stage-1 exploration report and return the exact seven-key schema for every model: file_path, file_ext, sep, comment_char, id_col, seq_col and prob_col. Output JSON only.

VERIFIED OUTPUT Observed schemas are persisted and reused through schema_memory.json

SOURCE `agents/runtime_prompts/data_analyst_extraction.md · run_meeting.py:330-331`

P08 Evaluation code generation and review

SYSTEM
Use only runtime-supplied model commands, dataset files, schemas and output contracts. Do not invent paths, dependency versions, model outputs, columns or metric values. Preserve missing predictions as missing.

USER
Generate one executable Python evaluation script and one launch script. A separate Data Architect reviews every referenced field; runtime code performs syntax, artifact, dependency, path and scientific metric validation.

VERIFIED OUTPUT Automatic HPC predictions and manual prediction tables converge at one evaluator

SOURCE `agents/runtime_prompts/coder.md · data_analyst_review.md · main.py · run_meeting.py`

P09 Manual prediction import contract

SYSTEM
Manual model outputs cannot bypass provenance or metric rules. Preserve the original file, align sequence IDs, labels and probabilities, and report duplicate, unmatched or invalid rows.

USER
Import the three supplied prediction tables into dataset-specific directories, standardize their schema and continue from the same scientific evaluator used by automatically executed models.

VERIFIED OUTPUT C_AMPs-predict test n=59,311 · Veltri test n=1,203 · ProteoGPT test n=1,796

SOURCE `import_manual_prediction_results.py · data/manual_predictions/ · data/results_manual/`

P10 Independent scientific result audit

SYSTEM
Interpret objective benchmark data without fabricating missing results. Distinguish execution failure from biological prediction performance and keep dataset-specific limitations visible.

USER
Review dynamic metric weights, real metric tables and quantitative scores. Explain performance under the meeting protocol, then write the model-level scientific judgement and evidence-limited ranking.

VERIFIED OUTPUT Three evaluation bundles produced for 18 usable models

SOURCE `agents/runtime_prompts/critic.md · scientific_evaluation.py · data/results_manual/*/eval_result.json`